Issue	**1**
Factory ID	**26347**
Date of Issue	**13-Apr-2026**

Select Enterprise

P-204, Arshad Town, Darbar Street Rajbah Road
Faisalabad
Punjab Pakistan

Latitude: [31.424016], longitude: [73.025898]
Date of inspection: 10-Mar-2026

Electrical Safety Inspection Report

Observations & Actions



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Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 10-03-2026 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

Contents

1. Executive Summary 5
2. Building Extents 6
   - 2.1 Building Use 8
   - 2.2 Utility Source 9
   - 2.3 Backup source 10
3. Documentation 11
4. Electrical Safety Measures 12
5. Remediation Support 13
   - 5.1 Engineering Support 13
6. Diagrams & Drawings 14
   - 6.1 Single Line Diagram (SLD) 14
   - 6.2 Electrical layout drawing 16
7. DBs & Elec Protection Systems 17
   - 7.1 Circuit breakers 17
   - 7.2 Hot Spots 19
   - 7.3 Earthing system 20
   - 7.4 Loop Connection 22
   - 7.5 Multiple Connections 24
   - 7.6 Openings 26
   - 7.7 Adequate accessibility 28
   - 7.8 Lint & dust inside panel board 30
   - 7.9 Lightning Protection System (LPS) 32

8. Fire Alarm & Detection System (FADS)..... 34
8.1 Non-functional or incomplete..... 34
9. Emergency Lighting..... 36
9.1 Provided but insufficient..... 36
9.2 Independent power source..... 38
10. Generator, Transformer & Sub-Station Room..... 39
10.1 Battery..... 39
11. Cabling & Wiring..... 41
11.1 Cable support and protection..... 41
11.2 Cable trench and channel cleanliness..... 43
11.3 Splicing of cables..... 45
11.4 Colour coding..... 48
11.5 Fixtures..... 50
12. Assessment Limitations and Assumptions..... 52

1. Executive Summary

On 10-Mar-2026, Syed Hassan Hussain and Zeeshan Haider on behalf of the Pakistan Accord carried out an initial electrical inspection for Select Enterprise, a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with factory management representative Mr. Muhammad Shabbir (Owner), Mr. Mohsin Siddique (HR manager), and other staff members of Select Enterprise.
The factory comprises of one main building owned by Select Enterprise. During the inspection, the area highlighted in blue in Figure 1.1 was inspected, with the site boundary indicated in red.

Main Building: A two-storey (Ground + Mezzanine +1) building used for solar panels, storage (roof), stitching, pressing, packing, meeting room (first floor), prayer area, storage (mezzanine), warehouse, store, offices and cutting (ground floor). The construction year of the building is 2017.

A summary of main concerns identified on-site during the assessment (non-exhaustive) that should be actioned is as follows:

Earthing system: The factory does not have an earthing system installed for the electrical installation. The absence of a proper earthing system poses a significant safety risk, including potential electric shock, fire hazards, and damage to electrical equipment.
Emergency Lights: Several areas lack sufficient emergency lighting and exit sign coverage, which may create hazardous conditions during an emergency. Poor illumination along walkways can increase the risk of trips and falls, while unclear or inadequately marked exit routes may cause confusion and panic among workers attempting to evacuate.
Fire Alarm and detection system: A manually operated hooter system was installed in the facility. A non-functional/incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm.

We see no reason to suspend operations in the facility due to these concerns, subject to the actions identified being addressed within the specified timeframes.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents



Figure 1.1. Site plan (Satellite view) - Red dashed line indicates site boundary highlighted (blue) area indicates building inspected.

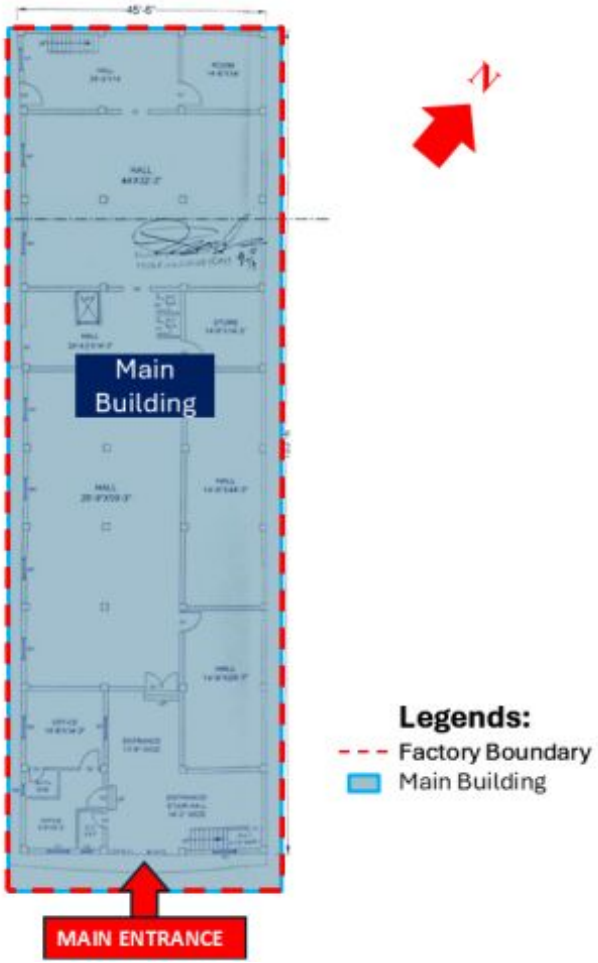


Figure 1.2. Site Layout of Select Enterprises

2.1 Building Use

The factory comprises one building and building use per floor is as follows:

L00- Ground floor is used as warehouse, store, offices, and cutting area.
M00- Mezzanine floor is used as prayer and storage area.
L01- First floor is used as stitching, pressing, packing, and meeting room.
L99- Roof is used for solar panels and storage.

2.2 Utility Source

The factory utilizes a low-voltage 400V connection from FESCO as its primary source. This connection is first terminated in their main distribution board on the ground floor. Further distribution is done from this panel throughout the facility. Further, a 30kW solar system is also installed in the factory, working as a primary source as well.

2.3 Backup source

This factory utilizes a 35kVA diesel generator as its backup source, placed inside the factory at ground floor. Emergency lighting has its own standalone battery backup.

3. Documentation

This table shows the details of the documentation provided by the factory.

S.NO	Description	Available (Y/N)	Created Date	Status / Remarks
1	Electrical Layout Drawing	No	-	-
2	Single Line Diagrams	No	-	-
3	Fire Alarm Drawing	No	-	-
4	Earth Test Report	No	-	-
5	Thermographic Test Report	No	-	-
6	Insulation Resistance Test Report	No	-	-
7	LPS Risk Assessments	No	-	-
8	Periodic Maintenance Record	No	-	-
9	Transformer Oil Testing Report	N/A	-	-

4. Electrical Safety Measures

During our electrical site inspection of Select Enterprise, a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within Select Enterprise. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a trained professional electrician.

Before commencing any installation and/or modification works, the proposed design must be submitted to the Pakistan Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Remediation Support

5.1 Engineering Support

Some items in the Corrective Action Plan (CAP) will require remediation by a qualified engineer. Please refer to the Remediation Plan for guidance on which tasks can be completed using internal resources and which require engineering support. F-202603-064118

Timeline	Remediation Plan
Medium Term CAP-0075184	Factories should begin remediation of CAP items such as cleaning, cable organisation, and basic corrections using internal resources. For systems or design-related remediation, engagement of a qualified Electrical Engineer is required.

6. Diagrams & Drawings

6.1 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the electrical single-line diagrams (SLDs) for the factory were missing. This deficiency was noted in all electrical rooms and areas where SLDs should be prominently displayed.

The factory lacks documented diagrams that illustrate the electrical distribution system, including the connection and configuration of all electrical components.

Single-line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202603-063811

Timeline	Remediation Plan
Medium Term CAP-0074613	Develop comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0074614	Implement a regular review and update process for the single line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

6.2 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the factory lacked electrical layout drawings - including small power layouts, lighting layouts, and containment layouts.

This absence was consistent across all areas where such documentation should be available.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202603-063819

Timeline	Remediation Plan
Medium Term CAP-0074636	Develop comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0074637	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

7. DBs & Elec Protection Systems

7.1 Circuit breakers

This section refers to the entire facility.

During the survey the circuit breakers installed throughout the facility inside the electrical panels and distribution boards were inspected.

It was observed that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes. This issue was observed in main distribution boards of both ground and first floor of the facility.

Specifically, some circuit breakers had higher ratings than the cables could safely handle, which poses a significant risk of overheating and potential fire hazards. Conversely, other circuit breakers had lower ratings than necessary, resulting in frequent and unnecessary tripping. To ensure both safety and operational efficiency, corrective action is required to align the circuit breaker ratings with the cable capacities. This adjustment will mitigate the risks associated with overheating and improve the reliability of the electrical system. F-202603-063816

Timeline	Remediation Plan
Medium Term CAP-0074626	Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.



Figure 2.1 - 225A circuit breaker on 25sq.mm cabe observed in main distribution board at ground floor

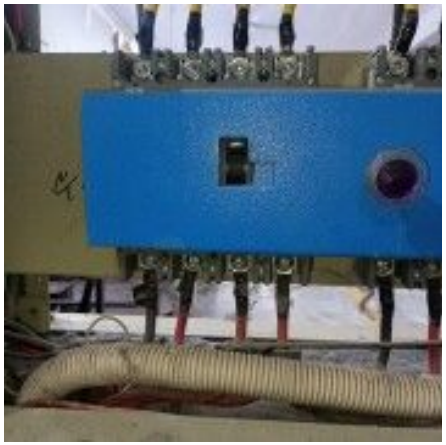


Figure 2.2 - 250A circuit breaker on 25sq.mm cable observed in main distribution board at first floor

7.2 Hot Spots

This relates to the entire facility.

During the inspection, Thermographic surveys were conducted on the electrical boards.

Circuit breakers are typically designed to operate safely within an ambient temperature range of -20°C to 40°C. Commonly used PVC-insulated cables typically have a maximum operating temperature of 70°C. Prolonged exposure to temperatures above these can degrade the insulation of the breaker as well as the cable and result in reduced performance.

It was observed that the thermography test report was not available at the time of inspection.

Hot spots in electrical systems pose significant life safety risks due to several factors. Overheated components can ignite surrounding materials, leading to electrical fires. This is particularly dangerous in densely packed panels and distribution boards where flammable materials may be present. Hot spots can degrade insulation on wires and components, increasing the risk of electrical shock to personnel who might come into contact with exposed conductive parts. Overheating can contribute to conditions that might trigger an arc flash, a dangerous high-temperature event that can cause severe injuries or fatalities to personnel working near the affected equipment. F-202603-063824

Timeline	Remediation Plan
Medium Term CAP-0074644	It is recommended that the factory conducts comprehensive thermographic surveys of all electrical panels to detect similar issues. Identified hot spots should be thoroughly reviewed to determine their underlying causes. Corrective actions must be taken to address and resolve these issues, ensuring safe and reliable operation of the electrical systems.
Longer Term CAP-0074645	Regular thermographic inspections should be incorporated into the maintenance schedule after every 6 months to proactively manage and mitigate the risks associated with overheating.

7.3 Earthing system

This relates to the entire facility.

During the inspection of the facility's electrical panels, distribution boards, and machinery, several observations with the earthing system were identified.

It was observed that the factory does not have an earthing (grounding) system installed. Neither an earth pit nor any earthing distribution network was established throughout the facility.

The lack of a proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202603-063814

Timeline	Remediation Plan
Medium Term CAP-0074621	i. Develop a detailed earth layout drawing for the factory and submit it to Pakistan Accord for review.
Medium Term CAP-0074622	ii. Install earthing systems where absent, ensuring compliance with BS 7671 18th Edition + A2 (2022) Chapter 54. Fit all electrical boards with the required earthing cables. Ensure that earthing conductors are properly connected and comply with BS 7671 18th Edition + A2 (2022) Section 543.1.
Medium Term CAP-0074623	iii. Implement equipotential bonding for all cable trays to comply with BS 7671 Regulation 411.3.1.2. Ensure all exposed conductive parts are bonded to the main earthing terminal.
Medium Term CAP-0074624	iv. Provide earth pit ohmic values test result as per IEC 61557 fall of potential method.
Longer Term CAP-0074625	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.3 - No earth connection observed in main distribution board at ground floor



Figure 2.4 - No earth connection observed in main distribution board at first floor

7.4 Loop Connection

This relates to the entire facility.

During the inspection, loop connections were identified within the electrical panels.

In these loop connections, one circuit breaker is energised by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system. This issue was observed at several locations, including the first floor of the facility.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202603-063829

Timeline	Remediation Plan
Medium Term CAP-0074674	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilised to provide each circuit with the required live, neutral, and earth connections.



Figure 2.5 - Loop connection observed in main distribution board at first floor of the facility

7.5 Multiple Connections

This relates to the whole facility.

During our inspection, multilooping was observed in a number of electrical panels across the facility.

In these cases, multiple circuits were being protected by a single circuit breaker. This issue was observed at several locations on ground floor of the building.

Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit. F-202603-063808

Timeline	Remediation Plan
Medium Term CAP-0074604	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022)



Figure 2.6 - Multiple connections observed in main distribution board at ground floor



Figure 2.7 - Multiple connections observed in genrator distribution board at ground floor



Figure 2.8 - Multiple connections observed in warehouse distribution board at ground floor

7.6 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

It was observed that unsealed openings were discovered in the panels and distribution boards, including the lift panel on the roof.

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202603-063830

Timeline	Remediation Plan
Medium Term CAP-0074677	Corrective actions are required to seal all openings and replace missing panels to restore the integrity and safety of the electrical systems. The electrical system shall ensure compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term CAP-0074678	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.9 - Unseald opening observed in lift panel at roof

7.7 Adequate accessibility

This relates to the entire facility.

During the visit, the electrical panels and distribution boards were examined for adequate accessibility.

It was observed that the changeover panel near generator area on the ground floor does not possess the required space for inspection or easy access to the panels and their components.

In several instances, the placement of electrical panels adjacent to machines that continuously reject heat has caused the surrounding temperature to exceed ambient levels by several degrees. This temperature rise further increases the temperature of the panels and boards nearby.

Limited accessibility to distribution panels and machine panels hampers effective maintenance and inspection procedures. This can potentially cause delays in addressing emergencies and troubleshooting electrical problems. These accessibility challenges could compromise the operational reliability and safety of the premises, as prompt maintenance and emergency responses are hindered. F-202603-063820

Timeline	Remediation Plan
Medium Term CAP-0076029	Electrical equipment shall be arranged to provide sufficient space for the initial installation and later replacement of individual items of electrical equipment. Sufficient allowance shall also be made for sufficient space for the initial installation and later replacement of individual items of electrical equipment in accordance with Section 132.12 of BS7671 18th Edition + A2 (2022).



Figure 2.10 - Inadquate accessibility observed for changeover panel near generator area at ground floor

7.8 Lint & dust inside panel board

This relates to the entire facility.

During the inspection, the cleanliness of electrical panels, distribution boards and electrical containment was assessed.

The inspection revealed a significant accumulation of lint and dust in most of the distribution boards, panels and trunking systems. This issue was observed at several locations, including ground and first floor of the facility.

The presence of lint and dust inside distribution boards poses several hazards. Accumulated lint and dust can act as fuel for fires if exposed to electrical currents or sparks, significantly increasing the risk of electrical fires. The cooling capacity of electrical components can also be impaired by the presence of dust and lint, leading to overheating.

Additionally, the accumulation of dust and lint can obstruct proper operation and maintenance, thereby impairing the functionality of the electrical components.F-202603-063825

Timeline	Remediation Plan
Medium Term CAP-0074646	All electrical panels and trunking runs shall be inspected and cleaned of any dust or lint in accordance with Section 522.4.2 of BS7671 18th Edition + A2 (2022)
Longer Term CAP-0074647	A regular maintenance procedure shall be established to ensure routine inspection and cleaning of all electrical panels and trunking runs. This will prevent the accumulation of lint and dust, thereby maintaining the safety and efficiency of the electrical systems.



Figure 2.11 - Lint and dust observed in generator distribution board at ground floor



Figure 2.12 - Lint and dust observed in warehouse distribution board at ground floor



Figure 2.13 - Lint and dust observed in main distribution board at first floor

7.9 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

The following observations were identified with the current lightning protection system:

i. A lightning protection system was provided, however, there was no study available to confirm its adequacy.
ii. No test results were provided for the lightning protection system, making it difficult to verify its effectiveness and compliance with relevant standards.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202603-063876

Timeline	Remediation Plan
Medium Term CAP-0074697	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment in accordance with IEC-62305. This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term CAP-0074698	If the risk assessment concludes that an LPS is necessary, proceed with the design and installation of an appropriate system in compliance with relevant standards such as BS EN 62305 (Protection against Lightning).



Figure 2.14 - Lightning arrestor installed on rooftop

8. Fire Alarm & Detection System (FADS)

8.1 Non-functional or incomplete

This section refers to the entire facility.

During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

It was observed that no fire alarm and detection system was installed in the facility. However, a 220 VAC hooters with a push-start button and some standalone smoke detectors were installed, which do not comply with standards.

A non-functional/incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm.F-202603-063818

Timeline	Remediation Plan
Medium Term CAP-0074630	1. Submit design documentation for the new FADS system to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets, and product certifications of all listed equipment to show full compliance with the Fire Alarm and Detection System (FADS) with NFPA 72.
Longer Term CAP-0074631	2. Submit documentation to the Pakistan Accord showing that an L/C has been opened for the import of any components required for the FADS design submitted earlier.
Longer Term CAP-0074632	3. Submit documentation to the Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on-site for installation.
Longer Term CAP-0074633	4. Once installation of the FADS system is complete, the factory is to submit full "as-built" documentation of the system to the Pakistan Accord for review.
Longer Term CAP-0074634	5. Factory to advise Pakistan Accord when FADS installation is fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation.

Longer Term CAP-0074635	6. Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel, and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.



Figure 2.15 - 220 VAC hooters installed for the facility



Figure 2.16 - Standalone smoke detector installed for the facility

9. Emergency Lighting

9.1 Provided but insufficient

The section refers to the entire facility.

During the inspection the emergency lighting arrangement throughout the facility was assessed.

The following observations were made:

i. Emergency lighting and exit signage layout for the factory was missing.
ii. Several areas lack sufficient emergency lighting and exit signage coverage.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202603-063828

Timeline	Remediation Plan
Medium Term CAP-0074662	i. Conduct a comprehensive survey to identify areas for adequate emergency lighting and directional signage, prepare a detailed emergency . Submit emergency lights drawing of entire facility including calculations / simulation to Pakistan Accord for review that comply with Building Code of Pakistan Fire Safety Provisions 2016. The egres routes/means of escape must be approved prior to submission.
Medium Term CAP-0074663	ii. Install additional emergency luminaires and exit signage as per initial submitted approved drawing to meet the coverage requirements specified in the Building Code of Pakistan Fire Safety Provisions 2016, ensuring sufficient illumination for safe evacuation and submit emergency lighting layout to Pakistan Accord.
Longer Term CAP-0074665	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.17 - No emergency light observed near warehouse area on ground floor of building



Figure 2.18 - Lack of exit signage observed at first floor of the facility



Figure 2.19 - No emergency light observed on stair case

9.2 Independent power source

The section refers to the entire facility.

During the inspection, it was observed that the emergency lighting within the facility was not connected to a dedicated circuit throughout the facility.

Proper circuitry is crucial to ensure that emergency lighting is automatically activated when general lighting fails. This automatic activation is essential for maintaining sufficient illumination, allowing for the safe evacuation of occupants and preventing panic during power outages. Further, the emergency lights and exit signs be reconfigured to be connected on a dedicated single circuit, which must not include any other load except for the emergency lighting system According to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202603-063810

Timeline	Remediation Plan
Medium Term CAP-0074609	Reconfigure the emergency lighting circuitry to ensure that emergency lights activate immediately upon failure of general lighting power, and provide a dedicated circuit for the emergency lighting system, as required by the Building Code of Pakistan Fire Safety Provisions 2016.
Longer Term CAP-0074612	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.

10. Generator, Transformer & Sub-Station Room

10.1 Battery

During the inspection of the battery system, the battery cells, connections, and enclosures were thoroughly examined.

It was observed that several battery terminals were exposed and not properly insulated, increasing the risk of electrical shock or short circuits. This issue was observed at generator area and at first floor.

Proper monthly and annual maintenance of the battery system is crucial for fire safety. Spilled fluid can cause corrosion and fires, while bare terminals increase the risk of electrical shocks and short circuits, potentially igniting fires or explosions. Non-compliant battery placement can obstruct ventilation, leading to hazardous gas buildup and fire risks. Addressing these issues is essential to prevent fires and ensure battery system safety.F-202603-063831

Timeline	Remediation Plan
Medium Term CAP-0074679	Insulate exposed battery terminals using appropriate insulating materials to prevent electrical shock and short circuits. Conduct a thorough inspection of all connections to ensure that there are no immediate risks of exposure.
Longer Term CAP-0074680	Establish a routine inspection and maintenance protocol to regularly check and maintain proper insulation of all battery terminals.



Figure 2.20 - Exposed battery terminals observed in generator area at ground floor



Figure 2.21 - Exposed battery terminals observed at first floor production area

11. Cabling & Wiring

11.1 Cable support and protection

This section refers to the entire facility.

During the inspection of the entire facility, a visual survey of the cable containment system was conducted.

It was observed that proper supports were not provided for cables; they were laid directly on the ground and hung along the walls. Inadequate cable support results in sagging or stress on cables, which can lead to damage or failure. This issue was observed at first floor and rooftop of the building.F-202603-063813

Timeline	Remediation Plan
Medium Term CAP-0074619	Secure cables with appropriate supports and insulate any exposed connections to prevent sagging, stress, and potential hazards. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support and 526.5 for proper terminations)
Longer Term CAP-0074620	Implement a comprehensive cable management system with regular inspections to ensure cables remain properly supported and insulated over time.



Figure 2.22 - Cable support not provided at first floor near metal detector machine



Figure 2.23 - Cable support not provided for solar cables on roof

11.2 Cable trench and channel cleanliness

This related to the whole facility.

During the inspection of the facility, it was observed that the cleanliness of the cable trunking system had not been maintained. Numerous trunking systems were found to be filled with a significant amount of fluff and dust, which poses a serious fire risk. This issue was observed at severl locations including first floor trunking system.

Accumulated dust and fluff can ignite when exposed to sparks or high temperatures, potentially leading to a fire hazard. Maintaining a clean cable trunking system is essential to minimise fire risks associated with accumulated debris F-202603-063822

Timeline	Remediation Plan
Medium Term CAP-0074642	Clean all cable trunking systems to remove accumulated dust and fluff, ensuring that any potential fire hazards are mitigated.
Longer Term CAP-0074643	Implement a regular maintenance schedule that includes routine inspections and cleaning of the cable trunking systems to prevent future accumulation of dust and fluff, thereby maintaining a safe environment and complying with safety standards.



Figure 2.24 - Cable trunk accumulated with fluff at first floor

11.3 Splicing of cables

This relates to the whole facility.

During the inspection, the cable installation was visually surveyed throughout the facility.

The following observations were noted:

i. Cables terminations within panels and switchgears were made without cable lugs, resulting in broken strands and loose connections. This presents a significant fire risk due to the potential for electrical arcing and overheating, which can ignite nearby materials and cause a fire. Furthermore, loose connections can cause voltage drops and equipment malfunctions, posing a life safety hazard. This issue was observed at several locations, including ground and first floor of the facility.

ii. PVC tape joints were found in several locations - including within distribution boards - where cables were either damaged or two cables were being joined together. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical shorts and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks. This issue was observed at several locations, including ground and first floor of the facility.

For any future electrical work, the termination of stranded cables must be done with cable lugs and PVC tape joints shall not be utilised. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety. F-202603-063826

Timeline	Remediation Plan
Medium Term CAP-0074652	i. Install cable lugs on all improperly terminated cables within panels and switchgears to secure connections and prevent breaking strands and loose connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022)
Medium Term CAP-0074653	ii. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022). Replace all damaged cabling with new wiring.
Longer Term CAP-0074654	i. Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections.

Longer Term CAP-0074655	ii. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques.



Figure 2.25 - PVC Tape joints observed in main distribution board at ground floor



Figure 2.27 - PVC Tape joint observed in main distribution board at first floor



Figure 2.26 - Cable terminated without lugs in warehouse distribution board at ground floor

11.4 Colour coding

This relates to the whole facility.

During the inspection of several distribution boards, deviations from standard cable colour coding practices were identified. Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires. This issue was observed at several locations, including the main changeover panel at ground floor.

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding as prescribed by BS 7671. This will ensure accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202603-063809

Timeline	Remediation Plan
Medium Term CAP-0074607	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labelling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.
Longer Term CAP-0074608	Develop and enforce a strict protocol for cable colour coding to be followed for all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices. Additionally, periodic audits should be conducted to ensure ongoing compliance with the standardized colour coding system, thereby preventing future deviations and ensuring sustained safety and operational efficiency.



Figure 2.28 - Colour coding issue observed in main changeover panel at ground floor

11.5 Fixtures

This relates to the whole facility.

During the inspection, the electrical fixtures throughout the facility were visually surveyed.

The following observations were noted:

i. Loose lighting fixtures hung with cable tie: This poses a significant risk of equipment damage or electrical incidents due to the instability of the receptacles. Loose connections can lead to electrical arcing, which increases the risk of fire and can cause serious injury or equipment failure. This issue was observed at first floor of the building.
ii. Power receptacles mounted on combustible wooden ply: Installing receptacles on combustible materials like wooden ply increases the risk of fire if a short circuit or overheating occurs. Combustible materials can easily ignite, leading to potential fire hazards and endangering life safety. This issue was observed at ground and first floor of the facility.F-202603-063812

Timeline	Remediation Plan
Medium Term CAP-0074615	i. Secure the loose lighting fixtures with appropriate fixed supports to prevent movement and ensure stable connections.
Medium Term CAP-0074616	ii. Replace combustible wooden ply backings with non-combustible materials to reduce fire risk.
Longer Term CAP-0074617	i. Implement a regular inspection and maintenance schedule to check and secure all power receptacles, ensuring they remain safely and correctly installed.
Longer Term CAP-0074618	ii. Establish a facility-wide standard to use only non-combustible materials for mounting electrical components, and conduct regular audits for compliance.



Figure 2.29 - Wooden back supported fixture observed at ground floor



Figure 2.30 - Fixtures supported with cable tie at first floor



Figure 2.31 - Wooden back supported fixture observed at first floor for water cooler

12. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.